

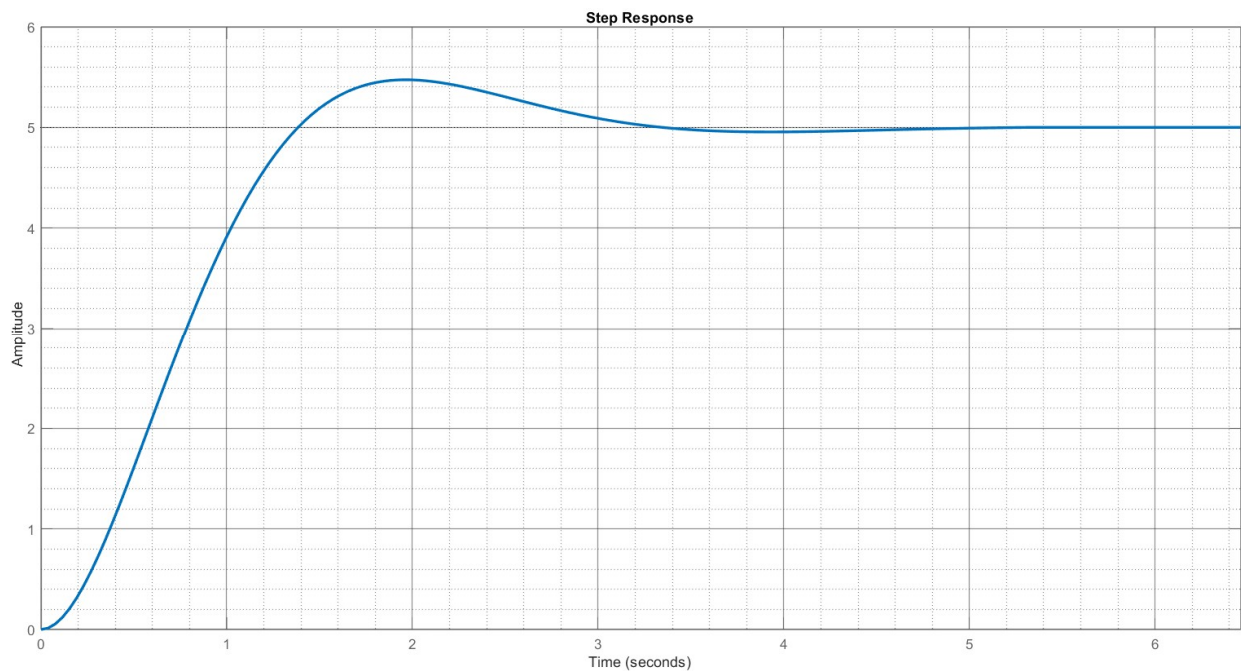
Name: _____

Student no: _____

Tut Test 2

The step response of a system is shown below. Based on your evaluation of the system's components, you think that it can be modelled as either a first-order system $G(s) = \frac{A}{\tau s + 1}$ or a second-order system $G(s) = \frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$.

- Explain which model you will be using, with a reason (2 marks), and hence
- Estimate the unknown values in the model based on the step response. (8 marks)



You may use the formulas below to help you:

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\cos \theta = \zeta$$

$$t_p = \frac{\pi}{\omega_d}$$

$$y_p = A \left(1 + e^{-\zeta\pi / \sqrt{1-\zeta^2}} \right)$$

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$t_{k\%} = \frac{\ln\left(\frac{k}{100}\sqrt{1-\zeta^2}\right)}{-\zeta\omega_n}$$

$$\%OS = 100e^{\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}\right)}$$

